



AI Basics, Ethics & Applications – Worksheet with Use Cases

Student Name & Roll#: _____

Grade/Class: _____

Date: _____

✨ Section A: AI, ML & DL Basics

1. Differentiate between AI, ML, and DL by matching them to their correct use cases. (3 Marks)

Term Use Case

- a. AI ___ 1. A smart assistant like Siri or Alexa
- b. ML ___ 2. Email filtering system that learns spam
- c. DL ___ 3. Facial recognition in smartphones

2. Define the following terms in your own words and give one example:

a. Machine Learning: _____

Example: _____

b. Deep Learning: _____

Example: _____

3. Common Data Terminologies – Fill in the blanks

- a. A _____ is a collection of structured or unstructured data.
- b. The _____ is what we are trying to predict in supervised learning.
- c. A _____ is a measurable input used to train a model.
- d. In AI, _____ data is used to train the model.
- e. _____ data is used to check how well the model performs.

🔄 Section B: AI Project Cycle

4. List the 6 stages of the AI Project Cycle.

- 1. _____
- 2. _____
- 3. _____
- 4. _____
- 5. _____
- 6. _____

5. Use Case Scenario: "AI in Education" – Match the stages to the steps.

A school uses AI to predict which students might need extra support based on exam scores and class participation data.

Step Description	AI Project Cycle Stage
a. Collecting past exam scores and attendance data	_____
b. Understanding how attendance relates to performance	_____
c. Using algorithms to build a predictive model	_____
d. Using the model to flag students for support	_____
e. Getting feedback from teachers and adjusting the model	_____
f. Monitoring how students improve after AI interventions	_____

** Section C: Domains of AI & Use Cases****6. Fill in the table with the correct AI domain and a matching use case: (9 Marks)**

Use Case	AI Domain
a. AI detects human emotions from voice	_____
b. AI chatbot answers customer questions	_____
c. AI forecasts crop yields using weather data	_____

** Section D: AI Ethics & Bioethics****7. Why do we need ethical frameworks in AI? Give two reasons with examples. (4 Marks)**

Reason 1: _____

Example: _____

Reason 2: _____

Example: _____

8. List three factors that influence our decision-making when creating AI systems. (3 Marks)

a. _____

b. _____

c. _____

9. Bioethics Principles – Match each principle with a use case: (4 Marks)**Bioethics Principle Use Case Example**

a. Autonomy ___ A doctor uses AI to recommend, but patient decides

b. Beneficence ___ AI detects cancer early to improve health outcomes

c. Non-maleficence ___ AI is tested to avoid harming patients

d. Justice ___ Ensuring rural patients also get fair AI access

10. Case Scenario:

An AI system recommends surgery for a patient but doesn't explain the reasons. The patient is confused and fearful.

Question: Which bioethics principles are not being followed? What should be improved?

Answer: _____

✓ Submission Instructions:

- Please complete the worksheet in neat handwriting.
- Use examples to support your answers where asked.
- Submit by: **[23/06/2025]**